

mance, low-power, 8-bit AVR microcontroller. It has 32 kB of in-system self-programmable flash, 2 kB of static random access memory (SRAM) and 1 kB of electrically erasable programmable read-only memory (EEPROM). Some of the on-chip peripheral features are:

1. Two 8-bit timers/counters with separate prescaler and compare modes
2. One 16-bit timer/counter with separate prescaler, comparator mode and capture mode
3. Four pulse-width-modulated (PWM) channels
4. 8-channel, 10-bit analogue-to-digital converter (ADC)

5. Byte-oriented, two-wire serial interface (TWI)

6. Programmable serial USART (universal synchronous asynchronous receiver transmitter)

7. Master/slave SPI serial interface

8. Programmable watchdog timer with separate on-chip oscillator

9. On-chip analogue comparator

The development board

This ATmega32 development board has the following features:

1. In-system programmable with an on-board serial-port-based programmer

2. Onboard regulated and filtered

power supply with power indicator LED

3. Eight LEDs, which

are selectable through DIP switches (provided on the board) for status or debugging purpose

4. Five tactile switches for external inputs or interrupts; a reset switch to safely reset the microcontroller externally, two switches on external interrupt pins and two more for general purpose

5. On-board LCD connector (which can be used for other applications too) along with a preset to control the contrast

6. On-board PC-MCU serial port link for use of the USART module in the MCU for two-way communication with the PC

7. On-board piezobuzzer selectable through a jumper for status or debugging purpose

8. On-board two-wire serial EEPROM of 4 kB (512x8) in addition to 1 kB of on-chip EEPROM

9. All the MCU ports are easily accessible through single-line male header (berg strip connectors)

10. Two 8-pin male header strip connectors for providing power supply to external devices and sensors

11. On-board variable voltage source using a preset, provided via a switch to PA7 for ADC test

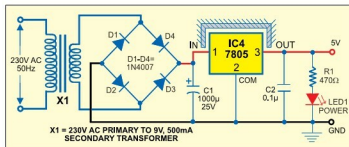


Fig. 2: Power supply circuit for ATmega32 development board

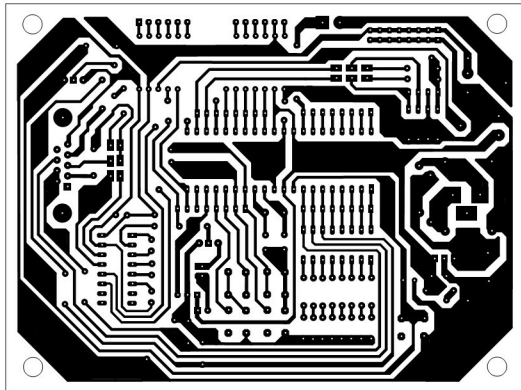


Fig. 3: An actual-size, single-side PCB for the ATmega32 development board

Circuit description

Fig. 1 shows the circuit of ATmega32 development board. At the heart of the circuit is the 40-pin ATmega32 MCU, which is connected to the board through a 40-pin dual in-line package (DIP) IC socket.

The MCU has four input/output (I/O) ports: Port A (pins 33 through 40), Port B (pins 1 through 8), Port C (pins 22 down through 29) and Port D (pins 14 through 21). Each of these port pins is made available through external male header strips. In order to draw